

## CPT306 Assignment 2: 3D Escape Room Detailed Specification

### 1. Game Overview: "The SIP Lab Lockdown"

The player awakens in a high-tech isolation chamber. To escape, they must restore power, solve a sequence of logical puzzles, and unlock the heavy security door.

**Theme & Aesthetics:** Sci-fi/Industrial. All environment elements should be built using Unity primitives (Cube, Cylinder, Sphere, etc.) with high-contrast materials to create a "clean but claustrophobic" atmosphere (If you have a better solution, you are welcome to come up with it freely).

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### 2. Environment & Level Design (The Grid & Room)

- **Room Dimensions:** A single primary room of **10m x 4m x 10m**.
  - **Boundaries:** The room must be fully enclosed by walls, a floor, and a ceiling.
  - **Start Position:** Player starts at coordinates (0, 1.1, -4).
  - **Exit:** A "Main Blast Door" located at the center of the North wall.
  - **Lighting (Atmosphere Requirement):**
    - **State A (Initial):** Low-intensity Red emergency lighting (using Point Lights).
    - **State B (Powered):** High-intensity Neutral White lighting (using Directional or Area Lights).
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### 3. Controls & Interaction (The Raycasting Requirement)

The game must use a **First-Person PC Control** scheme:

- **Movement:** W/A/S/D for translation.
- **View:** Mouse movement for 360-degree rotation.
- **Interaction (Raycast):**
  - A fixed crosshair in the center of the screen.
  - Interaction range: **3.0 units**.
  - Visual feedback: Display a text prompt (HUD) when the raycast hits an Interactable object (e.g., "Press E to Pick Up").

- Interaction Keys: E or Left Mouse Click.
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#### 4. Puzzle Progression & State Management

The game must implement a **Sequential State Machine** to handle the following 3 puzzles:

##### Puzzle 1: Power Restoration (Spatial/Physics Puzzle)

- **Requirement:** Find a "Energy Core" (Blue Cylinder) hidden behind physical obstacles.
- **Mechanism:** At least 5 "Crates" (Cubes with Rigidbody) must be piled in a corner. The player must physically move/push them to find the hidden Energy Core.
- **Trigger:** Interact with the "Generator Terminal".
- **Effect:** Global state `isPowerOn` becomes `True`. Lighting transitions from Red to White. Keypad (Puzzle 2) becomes active.

##### Puzzle 2: Keypad Override (Logical/Data Puzzle)

- **Requirement:** Input a 4-digit code to open a locked safe.
- **Logic:**
  - A visual clue (Text on a Plane) is hidden in the room: "ACCESS: 7294".
  - A 3D Keypad with 10 interactable buttons.
- **Constraint:** The Keypad only accepts input if `isPowerOn == True`.
- **Reward:** On correct input, a "Safe Door" rotates open, revealing the "Security Keycard".

##### Puzzle 3: The Final Escape (Inventory/Condition Puzzle)

- **Requirement:** Open the Main Blast Door.
- **Mechanism:** The door terminal checks for two conditions:
  - Condition 1: `isPowerOn == True`.
  - Condition 2: `hasKeycard` is in the Player's Inventory.
- **Effect:** The door slides open (Lerp/Animation). Walking through the door triggers the Victory Screen.

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## 5. User Interface (HUD & Menu)

- **HUD:**
  - Crosshair (Center Dot).
  - Interaction Prompt text (Center Bottom).
  - Inventory slot (Top Right, e.g., "Holding: Nothing" or "Holding: Keycard").
- **Menu Flow:**
  - **Start Screen:** Options to Start or Quit.
  - **Pause Menu (Esc or Space):** Options to Resume or Restart Level.
  - **Victory/Game Over Screen:** Clear messaging with time taken to escape.

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## 6. Implementation Constraints (Anti-Penalty Rules)

- **Collision:** Player and entities must not phase through walls.  
Use Rigidbody and CapsuleCollider for the player.
- **Cursor Management:** The mouse cursor must be locked during gameplay and unlocked/visible when a UI Menu is open.
- **Optimization:** Use a LayerMask for the Raycast to only detect interactable objects.
- **No External Assets:** All objects must be created via code or Unity primitives.